Juan Carlos Delgado-Pivaral

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EDUCATION:

Harvard College (GPA: 3.735) Expected May 2029

- Bachelor of Science (S.B.) Candidate in Mechanical Engineering

TECHNICAL SKILLS:

Software & Design: SolidWorks, AutoCAD, Onshape, Technical Drawings, Python, LaTeX, Design for Manufacturing
Manufacturing: CNC Machining, Manual Milling, Lathe, 3D Printing, Laser Cutting, Mold Fabrication, CAM
Languages: English (Native), Spanish (Native)

ENGINEERING EXPERIENCE:

Westin Automotive Products San Dimas, CA
Product Development Engineering Intern *June 2026 – August 2026*

- Designed automotive lighting assemblies in SolidWorks and managed their transition from prototype development to manufacturing release.
- Developed prototypes alongside machinists and conducted vehicle fit-ups to evaluate component integration, identify design issues, and refine mechanical designs.
- Coordinated directly with an overseas manufacturer to support product releases for production and prepared detailed technical documentation for mechanical component replacement.

Harvard Undergraduate Automotive Society – Formula SAE Cambridge, MA
Mechanical Team | Steering Sub-team *September 2025 – Present*

- Designed steering components and assemblies in SolidWorks, supporting vehicle development through CAD modeling, prototyping, and mechanical integration.
- Evaluating impact attenuator designs and developing an anti-intrusion plate mounting assembly in SolidWorks, considering chassis integration and Formula SAE requirements.

Harvard Undergraduate Robotics Club – University Rover Challenge Cambridge, MA
Mechanical Engineer | Science Team *September 2026 – Present*

- Serve as the designated mechanical engineer for both science subteams, translating scientific objectives into mechanical designs and coordinating system integration.
- Designing a rover-mounted drilling and auger system for subsurface soil collection, accounting for chassis geometry, mounting configurations, and required drilling depths.
- Designing a soil caching system to separate shallow and deep samples, incorporating removable storage and a retrieval mechanism meeting a five-minute requirement.

PROJECTS:

Battle Robot: Mechanical Design & Manufacturing

- Designed drivetrain, gearbox, wheel, and structural components in SolidWorks, incorporating manufacturing constraints, component integration, and mechanical assembly requirements.
- Manufactured and assembled custom robot components through machining, laser cutting, 3D printing, and molding, refining designs through prototyping, fit-ups, and mechanical testing.
- Collaborated on robot integration and performance improvements, earning 1st place in the final competition.

LEADERSHIP:

SHPE (Society of Hispanic Professional Engineers) Cambridge, MA
Co-Director of Outreach *September 2025 – Present*

- Lead chapter outreach through newsletters, social media, and event coordination, promoting professional opportunities and strengthening engagement with students and partner organizations.

ADDITIONAL EXPERIENCE:

Pauline's Food Service (Family-Run) Los Angeles, CA
Cook, Caterer, Translator *January 2020 – Present*

- Coordinate catering for 40–500+ guests, provide bilingual translation, and trained 5+ new employees.

Covina High School Covina, CA
Multi-Subject Tutor *August 2024 – July 2025*

- Tutored students in mathematics through Calculus BC, physics, and chemistry in individual and group sessions.

EXTRACURRICULAR INVOLVEMENT:

Harvard Men's Club Soccer Cambridge, MA

- Selected to Harvard Men's Club Soccer A Team, competing in intercollegiate matches and tournaments.